Abstract

Drug discovery is a lengthy, expensive, and risky process. It may potentially be possible to make the process less expensive and shorter by implementing drug repurposing, which refers to searching for alternative therapeutic uses of approved drugs. As approved drugs have undergone safety tests, drug repurposing could help speed up the development of novel treatments.

The objective of the current project is to conduct a comparison between different supervised machine learning models that predict the likelihood of discovering new uses of existing approved drugs based on public biomedical data sets. Data related to drug–disease associations and the overall properties of drugs will be obtained from biomedical resources such as the DrugBank database. This problem would be formulated as a binary classification task.

Logistic Regression, Random Forest, and Support Vector Machine will be used as supervised machine learning algorithms. They will be compared in terms of their accuracy, precision, recall, F1 score, and ROC-AUC.

The anticipated results of the proposed study will be the determination of machine learning approaches that work better in the area of computational drug repurposing. The study will show how machine learning can assist researchers in making drug discovery a more efficient process.

Introduction and Background

Drug discovery is one of the most crucial stages in the field of health care and pharmaceuticals. Nonetheless, drug discovery from scratch may be highly expensive and lengthy. For instance, developing and commercializing a new drug may take up to ten years and even billions of dollars. Besides, most drug candidates fail during their trials due to toxicity or inefficiency. Therefore, there are efforts in the current era to discover drugs using alternative methods which will not require a lot of time and money. One such method is drug repurposing or repositioning.

Drug repurposing is an activity which involves finding new indications for already approved drugs. The main advantage of repurposing drugs is that already approved drugs have undergone a number of phases regarding safety and efficacy. Thus, these drugs can be reused to treat other illnesses much more conveniently and cheaply than creating new medicines. Over the last few years, drug repurposing has gained popularity owing to the rise in biomedical information and computational tools. As recent research has proved, it is possible to detect the unknown connections among medicines, diseases, genes, and biological targets through computational methods.

The fast development of biomedical and public health-related data and databases has led to the emergence of new opportunities for implementing Machine Learning approaches into drug repurposing research. For instance, the DrugBank, ChEMBL, PubChem, among others, contain much useful information on drugs, compounds, biological targets, drug-disease interactions, etc. Machine learning techniques are capable of analyzing such big volumes of data and detecting some regularities which would otherwise remain unnoticed.

A variety of machine learning methods are used to solve computational drug repurposing problems, including Logistic Regression, Random Forest, Support Vector Machine (SVM), deep learning techniques, and others. The most popular approach is the application of supervised machine learning, since it allows for detecting patterns from labeled data and predicting new potential drug-disease connections. It has been proved by several recent researches that Random Forest and SVM algorithms are highly efficient in drug repurposing.

Even with recent advancements, there are still some issues that remain unresolved in computational drug repurposing. Datasets used in computational biology are often huge and unstructured, as well as noisy and imbalanced. Algorithms based on machine learning can behave differently with different datasets, feature representations, and validation techniques. Furthermore, many existing studies rely on sophisticated techniques such as deep learning or networks, which require both heavy computational machinery and extensive knowledge of biological science. Thus, there is still room for comparative studies that would explore the utility of supervised machine learning algorithms with public biomedical datasets.

The primary stakeholders in this study are pharmaceutical firms, medical researchers, and biomedical scientists, along with patients. Pharmaceuticals can benefit from a faster and cheaper way of discovering new drugs, whereas researchers can get a better grasp of machine learning techniques and how to use them in drug repurposing processes. Ultimately, patients can benefit from a quicker identification of drug repurposing opportunities. In the long term, patients may benefit from faster identification of potential treatments for diseases.

This project aims to perform a comparative analysis of supervised machine learning algorithms for predicting potential new therapeutic applications of existing approved drugs. Biomedical datasets from public sources will be used in order to generate predictions, which will then be evaluated for performance using various machine learning techniques. This project is designed to contribute to current research into computational drug repurposing through analysis of the advantages and disadvantages of machine learning techniques.

Aims and Objectives

Aim:

The main aim of this project is to evaluate and compare supervised machine learning algorithms for predicting potential new therapeutic applications of existing approved drugs using publicly available biomedical datasets.

Objectives:

1. To review existing literature related to drug repurposing and machine learning techniques used in computational drug discovery.
2. To collect drug–disease association data and drug-related information from publicly available biomedical databases such as DrugBank and other related resources.
3. To preprocess and prepare the collected biomedical data for machine learning analysis by handling missing values, formatting data, and selecting relevant features.
4. To formulate the prediction of drug–disease associations as a supervised binary classification problem.
5. To implement and train supervised machine learning models including Logistic Regression, Random Forest, and Support Vector Machine (SVM).
6. To evaluate and compare the performance of the machine learning models using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
7. To analyse the strengths, limitations, and effectiveness of different machine learning algorithms in computational drug repurposing tasks.
8. To identify possible future improvements and research directions for machine learning-based drug repurposing systems.

Expected Deliverables

1. The expected deliverables of this project include:
2. A cleaned and preprocessed biomedical dataset suitable for machine learning analysis.
3. Implementation of multiple supervised machine learning models.
4. Comparative performance analysis of the implemented models.
5. Visualisations and evaluation results for model comparison.
6. A final dissertation report discussing the methodology, results, limitations, and conclusions of the study.

Intermediate Deliverables

During the development of the project, several intermediate deliverables will also be produced, including:

1. Literature review summary
2. Dataset collection and preprocessing report
3. Initial exploratory data analysis results
4. Preliminary model implementation and testing
5. Evaluation and comparison results of machine learning models

These intermediate deliverables will help track the progress of the project and support the successful completion of the final dissertation.

Literature Review

Drug repurposing has emerged as a promising field of research during the past few years due to the limitations associated with conventional methods of drug discovery that involve extremely high costs, long time, and an extremely high rate of failure. Drug discovery using conventional methods may take from 10 to 17 years with expenses ranging into billions of dollars. Drug repurposing is all about identifying novel applications of already existing drugs, making the process much faster, less costly, and less risky considering that such drugs' safety and toxicity profile has been established already.

The emergence of recent advances in biomedicine database resources, big data tools, and computational strategies has resulted in greater application of Machine Learning in drug repositioning studies. There is a vast amount of data available from public biomedicine databases such as DrugBank, PubChem, ChEMBL, and BindingDB, which include knowledge on drugs, molecular structures, biological targets, pathways, and links between drugs and diseases. This data can be processed by machine learning algorithms for predictive analysis.

Various computing techniques that were applied in drug repurposing include phenotypic screening, target-based techniques, network-based techniques, molecular docking techniques, and machine learning techniques. These computing techniques for drug repurposing include the application of various supervised machine learning techniques which use labeled biomedical data and predict potential drug disease interactions. Conventional supervised machine learning techniques like Logistic Regression, Random Forests, and Support Vector Machine (SVM) are popular for their efficiency and ability to deal with biomedical data.

Recent literature indicates that Random Forest models usually exhibit excellent predictive performance in computational drug repurposing studies since they can efficiently manage nonlinearity, high dimensionality, and noise in biomedical data. In a study on machine learning-based drug repositioning, it was revealed that Random Forest models outperformed other models in terms of predictive power in drug–disease association prediction based on biomedical network datasets. It was found that machine learning approaches were capable of accurately identifying repurposing drugs through their similarity, disease associations, and biological networks.

Support Vector Machine (SVM) models have also been widely applied in computational drug repurposing research. In one such recent application on colorectal cancer, SVM classifiers

using various molecular descriptors, feature selection techniques, and cross-validation strategies were employed to identify possible drug candidates. The study demonstrated the importance of data pre-processing, descriptors generation, and feature engineering in achieving better performance. Support Vector Machine (SVM) classifiers are highly effective for biomedical classification tasks, particularly because they can efficiently manage high-dimensional feature spaces.

A significant study conducted another predictive modeling approach where a Random Forest model was used to predict SARS-CoV-2 main protease binding affinity using FDA approved drugs. Molecular descriptors, balanced data sets, repeated cross-validation and prediction confidence analysis were some of the techniques applied to predict COVID-19 drug repurposing targets. This is an example of how machine learning could play a critical role in identifying therapeutic drugs within a relatively short time period at minimum cost when there is a global health crisis.

Besides conventional supervised learning techniques, recent researches have considered the utilization of deep learning, graph-based learning, network pharmacology, and drug repurposing based on structure. Deep learning and graph neural network algorithms have proven to be highly effective in predicting drug–target interactions and other complicated biological associations. Moreover, AI-based researches have indicated that the significance of artificial intelligence is increasing for the purpose of rapid drug repurposing and enhancing predictive modeling. Nevertheless, advanced deep learning methods need massive data sets, extensive computing power, and substantial biological knowledge.

In contrast to the sophisticated deep learning algorithms, the conventional supervised machine learning models are more than adequate for this study because they are computationally feasible, relatively interpretable, and applicable to structured biomedical data sets. However, there are certain obstacles that exist within the realm of computational drug repositioning studies. First, biomedical data sets are prone to noise, incompleteness, imbalance, and heterogeneity. Moreover, since various experiments employ dissimilar data sets, pre-processing techniques, feature engineering schemes, and performance evaluation methods, it becomes challenging to conduct model comparison. Furthermore, certain machine learning models tend to function as black box systems, where it becomes difficult to explain their predictions in health care settings.

Although many machine learning and AI-based approaches have been proposed for computational drug repurposing, there is still a need for systematic comparative evaluation of traditional supervised machine learning algorithms using publicly available biomedical datasets and standard evaluation metrics. This research will provide an opportunity to analyze the performance and advantages and disadvantages of various supervised machine learning models in the prediction of new therapeutic uses for existing approved drugs.

Technologies and Resources

This project will use publicly available biomedical datasets and machine learning technologies to predict potential new therapeutic applications of existing drugs. The project mainly focuses on supervised machine learning methods and data analysis techniques.

Technologies

The main programming language used in this project will be Python because it is widely used in machine learning, data science, and biomedical research. Python provides many libraries and tools that support data preprocessing, model development, and evaluation.

Several machine learning and data analysis libraries will be used, including:

1. scikit-learn for implementing machine learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine (SVM).
2. pandas for data handling and preprocessing.
3. NumPy for numerical computation.
4. Matplotlib and Seaborn for data visualisation and plotting evaluation results.
5. Jupyter Notebook or Google Colab for model development, experimentation, and documentation.

The project may also use molecular descriptor tools and biomedical databases to obtain drug-related features and chemical information. Public biomedical datasets and resources will be collected from platforms such as:

1. DrugBank
2. PubChem
3. ChEMBL
4. BindingDB

These databases contain information related to drugs, molecular structures, drug–target interactions, biological pathways, and drug–disease associations.

Resources

The project will mainly require:

1. A personal computer or laptop for development and experimentation.
2. Internet access for downloading biomedical datasets and research papers.
3. Access to Kingston University online library resources and academic journals.
4. Cloud-based computational platforms such as Google Colab if additional computational resources are required.

Most of the software and datasets used in this project are publicly available and free for academic research purposes.

Alternative Technologies

Advanced approaches such as deep learning, graph neural networks, and molecular docking were also considered during the literature review. However, these methods often require large computational resources, highly complex datasets, and advanced biological knowledge. Since this project focuses on comparative analysis of supervised machine learning algorithms using structured biomedical datasets, traditional machine learning methods were considered more suitable, interpretable, and feasible for MSc-level research.

The selected technologies are appropriate because they are widely used in biomedical machine learning research and provide sufficient tools for implementing, evaluating, and comparing predictive models for drug repurposing tasks.

Method and Work Plan

This project will follow a supervised machine learning approach to predict possible new therapeutic applications of existing approved drugs using biomedical data. The overall workflow of the project includes data collection, preprocessing, feature extraction, model development, evaluation, and analysis.

Research Method

The project will begin with a review of existing literature related to drug repurposing, biomedical datasets, and machine learning techniques used in computational drug discovery. This will help identify commonly used methods, datasets, evaluation metrics, and current challenges in the research area.

After the literature review, publicly available biomedical datasets will be collected from sources such as DrugBank, PubChem, and other related databases. The collected data may include information related to drugs, chemical structures, molecular descriptors, and drug–disease associations.

The collected data will then be preprocessed before machine learning analysis. Data preprocessing may include:

1. removing duplicate records,

2. handling missing values,

3. formatting datasets,

4. feature selection,

5. normalization of numerical values.

The prediction task will be formulated as a supervised binary classification problem, where the machine learning models will predict whether a drug may be associated with a particular disease or not.

Several supervised machine learning algorithms will be implemented, including:

1. Logistic Regression,
2. Random Forest,
3. Support Vector Machine (SVM).

The dataset will be divided into training and testing sets. Cross-validation techniques may also be used to improve model reliability and reduce overfitting. The performance of the models will be evaluated using standard classification metrics such as:

1. Accuracy,
2. Precision,
3. Recall,
4. F1-score,
5. ROC-AUC,
6. Confusion Matrix analysis.

The final stage of the project will involve comparing the performance of the models, analysing their strengths and limitations, and discussing possible improvements for future research.

Proposed Workflow

The overall workflow of the project is expected to follow these stages:

Data Collection à Data Preprocessing à Feature Engineering à Model Training à Model Evaluation à Comparative Analysis à Final Report Preparation

Work Plan and Timeline

The project is expected to be completed in several phases over the dissertation period.

Phase 1 - Literature review and topic refinement (May – June)

Phase 2 - Dataset collection and preprocessing (June – July)

Phase 3 - Exploratory data analysis and feature preparation (July)

Phase 4 - Machine learning model implementation (July – August)

Phase 5 - Model evaluation and comparative analysis (August)

Phase 6 - Result interpretation and discussion (August – September)

Phase 7 - Dissertation writing and final revision (September)

Milestones and Deliverables

The major milestones of the project include:

1. Completion of literature review
2. Collection and preprocessing of biomedical datasets
3. Implementation of machine learning models
4. Completion of model evaluation and comparison
5. Preparation of dissertation draft
6. Final dissertation submission

The expected deliverables include:

1. Cleaned and processed dataset
2. Machine learning implementation code
3. Evaluation results and visualisations
4. Comparative analysis report
5. Final MSc dissertation document

Risk Analysis and Contingency Planning

Several challenges may arise during the project development process.

Poor data quality or missing data	Reduced model performance	Apply preprocessing and data cleaning methods
Imbalanced datasets	Biased predictions	Use balancing techniques and evaluation metrics such as ROC-AUC
Computational limitations	Slow model training	Use Google Colab or reduce feature complexity
Difficulty in feature selection	Reduced prediction accuracy	Use feature engineering and literature-guided selection
Time management issues	Delay in project completion	Follow planned schedule and milestones

By following a structured workflow and contingency planning approach, the project aims to remain feasible and achievable within the MSc dissertation timeline.

Ethics, Legal, Data Protection and Safety Considerations

This project mainly uses publicly available biomedical datasets and computational machine learning methods. No human participants, patient interviews, or clinical experiments will be involved in this research. Therefore, the ethical risks associated with the project are expected to be low.

The datasets used in this project will be collected from publicly accessible biomedical databases such as DrugBank, PubChem, and related academic resources. These databases mainly contain information related to drugs, chemical compounds, biological targets, and drug–disease associations. The project will not use confidential medical records or personally identifiable patient information.

Although the project does not directly involve human subjects, ethical responsibility is still important when applying Machine Learning techniques in healthcare-related research. Machine learning predictions may contain errors, bias, or uncertainty depending on the quality and balance of the dataset. Therefore, the results of this project will be considered research-oriented predictions only and not medical or clinical recommendations. Any predicted drug–disease relationships would require further biological and clinical validation before real-world medical use.

The project will also consider data protection and legal responsibilities. Since publicly available datasets will be used, the project is expected to comply with general research data usage policies. No personal or sensitive healthcare information will be collected or stored during the research process. The project will also follow academic integrity guidelines by properly citing datasets, research papers, and software tools used throughout the study.

Several challenges related to fairness and reliability may exist in biomedical machine learning research. For example, imbalanced datasets or incomplete biomedical information may affect model performance and prediction accuracy. To reduce these issues, appropriate preprocessing, validation methods, and evaluation metrics will be applied during model development.

From a safety perspective, this project does not involve laboratory experiments, chemical testing, or physical risks. All experiments will be computational and performed using software tools and publicly available datasets. The project therefore presents minimal physical safety risks.

Overall, the ethical, legal, data protection, and safety risks associated with this project are expected to be low. However, responsible use of biomedical data, transparent reporting of results, and acknowledgement of model limitations will remain important throughout the research process.

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